

Sean Luka Girgis

Senior AWS Data Engineer | Lakehouse, Glue/PySpark, SQL & AI-Assisted Delivery

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Professional Summary

Senior AWS Data Engineer with 20+ years of enterprise technology experience and strong hands-on background in AWS data platforms, Python, SQL, PySpark/Spark, AWS Glue, S3 lakehouse patterns, Athena, Redshift, ETL/ELT pipelines, data quality, metadata documentation, production troubleshooting, and AI-assisted engineering workflows. Experienced designing scalable data workflows, validating architecture decisions through implementation, optimizing SQL and pipeline performance, and delivering governed reporting datasets for regulated enterprise environments. Best aligned to AWS lakehouse roles requiring Glue/PySpark, S3, SQL, pipeline reliability, DQ/metadata patterns, CI/CD-aware delivery, and practical AI acceleration; CDK, dbt Core, Lake Formation, and event-driven AWS services are positioned as adjacent/active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, analytics, and forecasting workloads.
- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Migrated heavy forecasting and analytics workloads from on-prem Oracle patterns toward AWS/Redshift-backed processing, improving scalability and enabling cloud-based reporting workflows.
- Developed S3/Glue/Athena/Redshift-oriented lakehouse patterns using Parquet, partition-aware design, schema discipline, SQL optimization, validation checkpoints, and documented operational support practices.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve trust in pipeline outputs.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and historical retention models for performance, maintainability, and cost-aware access.
- Partnered with infrastructure, analytics, application, and business stakeholders to validate data architecture decisions, refine reporting requirements, and translate design into production-ready pipeline behavior.
- Used AI-assisted development workflows to accelerate Python pipeline work, documentation, quality review, JSON artifact generation, and repeatable delivery practices while preserving engineering review discipline.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and reporting processes for telecom-scale transaction systems.
- Worked with XML, SQL, Oracle, PL/SQL, messaging, and integration interfaces supporting billing, customer-management, and operational reporting workflows.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built automation scripts in Korn shell, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, and reporting access.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

AI-Powered Job Search Pipeline

Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.

Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, JSON, Quality Gates

HorizonScale — AI Capacity Forecasting Engine

Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.

Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation